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Suspicious online product reviews: An empirical analysis of brand and product characteristics using Amazon data

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ABSTRACT

Concerns over the authenticity of reviews hinder their usefulness. Consumers discount the information they get from reviews, and this particularly harms brands with high ratings. The authors argue that perceived brand strength, brand advertising effort, price, and sales each act as signals that help positive reviews of a brand seem more reliable. They investigate this by studying Amazon.com reviews of branded products from 16 product categories that have the resources and potential desire to advertise. They examine consumer perceptions of review authenticity as perceived by machine learning algorithms trained on human subjects, as well as by direct perceptions of human subjects during validation. The results indicate that having a strong brand and investment in brand advertising, along with price and sales rank, are valuable to managers since they form a certain protection from suspiciousness.

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1. Introduction

Online product reviews are a major source of information for consumers making purchase decisions. Shoppers who say that they read online reviews have increased from two-thirds (Wall Street Journal, 2016) to over three-quarters in recent years (BrightLocal, 2023). Opinions posted online that are based on consumers' experiences with products or services impact future consumer decisions. From a firm's perspective, online reviews of its products or services are critical because of their financial implications for sales (Floyd et al., 2014; Ho-Dac, Stephen, & Moore, 2013; Rosario et al., 2016; Sun, 2012; Zhu & Zhang, 2010).

Although consumer reviews are important and influence consumers' decision-making, consumers are often suspicious of their authenticity. According to BrightLocal's Consumer Review Survey[1] (2023), only 46% of respondents stated that they trust online reviews as much as personal recommendations from friends or family. The survey also revealed that 62% of consumers believed that they had read a fake review in the last year, with Amazon being the most frequently cited source. The authenticity of reviews is crucial for consumers to trust the ones they read. The 2022 BrightLocal Consumer Review Survey asked, "Does the existence of fake reviews and fake reviewers make you distrustful of online reviews in general?" Over 60% of the respondents answered "Yes".

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[1] A total of 1,117 US-based consumers from BrightLocal's Local Consumer Panel surveyed in January 2023.

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Review fraud is neither a hypothetical nor a rare practice. Approximately 5% of reviews on a retailer's website without a verified purchase feature are created by customers with no record of purchasing the product (Anderson & Simester, 2014). A server breach in May 2021 in China revealed that over 75,000 Amazon seller accounts had been buying customer reviews. On an average day, there are over 20 Facebook groups seeking to buy realistic, seemingly five-star reviews (He, Hollenbeck, & Proserpio, 2022). In a modern competitive business environment, sellers may be tempted to perpetrate review fraud by creating positive reviews of their products and/or denigrating reviews of competitors' products, especially when they believe that their manipulated reviews are undetectable by customers.

The potential harm from review manipulation affects sellers, brands, and opinion-sharing platforms. Customers who purchase a product based on promotional reviews and then find a gap between the product's expected and actual performance may make suboptimal choices in the future (Mayzlin, Dover, & Chevalier, 2014). Research has shown that even a small number of fake reviews can significantly affect the visible status of a business, which is its position on an e-commerce platform (He et al., 2022; Lappas, Sabnis & Valkanas, 2016).

Harm can occur just from the suspicion of review manipulation. Products with suspected fake reviews are believed to be of lower quality, leading to a decrease in sales (Zhang, Cui, & Peng, 2018). Experimental results show that suspiciousness negatively affects attitudes toward the brand and hurts the reputation (and thus the revenue) of the recommendation systems of opinion-sharing platforms (Harrison-Walker & Jiang, 2023). Once customers begin to mistrust the reviews on a platform, they may give reviews posted there less weight, even though most of them are created by genuine customers (Munzel, 2015).

Although marketing scholars have examined fake reviews (Luca & Zervas, 2016; Mayzlin et al., 2014; Moon, Kim, & Iacobucci, 2021), few articles have examined the concept of *perceived* fake reviews, despite their importance in leading consumers' feelings, opinions, and subsequent purchasing decisions. We study deception in online product reviews as perceived by consumers with machine learning algorithms trained on human subjects, as well as with direct perceptions of human subjects in validation processes. Unless the author admits it, no one, including the platform, can know for certain which reviews are fake. As such, our study is related to the marketing literature on the perceived deception of buyers in advertising and salesperson communications contexts. Our goal is to present a novel exploration of reviews that are considered suspicious. We seek to identify which brands are more prone to fall into being perceived as having suspicious reviews, as well as which reviews are more inclined to be categorized as suspicious.

We focus on the following key research question: Are there any characteristics of a product/brand that consumers/companies can use to predict the likelihood of a suspicious review? Our main objectives are to (1) determine if perceived brand strength and/or brand advertising effort are predictive of which reviews consumers/companies will deem as suspicious, (2) examine if a product's price and/or category sales rank are more likely to yield reviews that consumers consider suspicious, and (3) investigate contextual conditions that may give rise to these effects.

We theorize four marketplace signals—perceived brand strength, brand advertising effort, price, and sales rank—that help positive reviews seem more reliable. We tested these using observational data from Amazon.com. We found that lower perceived brand strength and lower brand advertising efforts are associated with a brand's online reviews being more likely to be perceived as suspicious by consumers. As price increases and category sales rank is closer to #1, a brand's reviews are less likely to be perceived as suspicious. In addition, all else being equal, technology products have a lower likelihood of suspicious reviews. The perception of suspicious reviews can also depend on the post (e.g., too positive or usage of certain language), which is not the focus of this study. Variables that describe the post are included as controls.

Our results have several managerial and academic implications. First, the paper examines perceived fake reviews, which we call suspicious reviews.[2] With few exceptions, past research has investigated fake reviews but has not addressed deception in online product reviews as *perceived* by consumers, even though perceived fake reviews are influential in affecting consumers' attitudes and behaviors and ultimately sales. We highlight the importance of brand strength and brand advertising investments, which can help protect brands from being perceived as untrustworthy by consumers.

Second, our study identifies the characteristics of suspicious online reviews that could help consumers become more knowledgeable and wary of potentially manipulative reviews. Reviews from weaker brands, brands with lower advertising effort, and reviews for lower-involvement products are more likely to be suspicious. We hope that consumers will become more empowered and sensibly skeptical, leading to a more trustworthy and legitimate online business environment. Finally, this paper examines several contextual factors that provide actionable insights for interested parties, such as e-commerce platform managers and general consumers.

2. Research background and conceptual framework

Under conditions of asymmetric information, a signal is an observable element or action that credibly communicates the level of some unobservable element (c.f., Kirmani & Rao, 2000). A common context in marketing is that consumers rely on elements that include brand name, advertising, product features, and price to gauge quality across competitive products

[2] We define a "suspicious review" as *perceived* by consumers. This is different from Schuckert, Lui, and Law (2016) who used the term to refer to a gap between the overall rating and the mean rating over five service attributes for hotels on TripAdvisor. We also differ from the eWOM skepticism construct developed by Zhang, Mo, & Carpenter (2016), which measures an individual's suspicions/distrust towards all eWOM messages.

under uncertainty about their performance (Dawar & Parker, 1994). In our context, consumers are faced with uncertainty about the incidence of manipulative online reviews for a brand. We posit that perceived brand strength, brand advertising effort, price, and sales rank can serve as signals that help allay suspicions about the authenticity of a brand's online reviews.

2.1. Brand strength

Brand differentiation and brand relevance together determine a brand's strength (see Lovett, Peres, & Shachar, 2014; Mizik & Jacobson, 2008). A strong brand is both differentiated in the marketplace and relevant to consumers.

Spence (1973) argued that for an element or action to be a signaling device, it should be costly to adopt, and such costs are higher for "bad" sellers than for "good" sellers. Then, in equilibrium, buyers correctly infer the sellers' true type based on their signaling strategies. A brand must devote time and effort to monitoring and building up a strong brand and receiving strong positive feedback from buyers after each transaction (Li, Srinivasan, & Sun, 2009). The signaling costs are related to the future revenue at stake, and a low credibility brand must incur more costs to attain the same brand strength, so signaling costs are greater, thus satisfying both of Spence's (1973) conditions.

In addition, sellers with relatively weaker brand strength commit review fraud more frequently because they have a stronger economic incentive to gain from it, compared to sellers with a stronger brand. Local and independent hotels (Mayzlin et al., 2014) and restaurants (Luca & Zervas, 2016) commit review manipulation more frequently than national chains. Less-known and lower-quality brands are almost exclusively the ones observed buying fake reviews on Facebook (He et al., 2022). Furthermore, manufacturers of products with strong brands may be expected to monitor their sellers, which acts as an added deterrent to sellers of those brands creating manipulative reviews.

Collectively, signaling, economic incentives, and monitoring by the brand owner all predict that a review of a stronger brand is less likely to be perceived as suspicious. The findings from single-category studies of non-contractual services should remain valid in our product context, where we examine multiple product categories and where the seller is often not the brand owner.

2.2. Brand advertising effort

A brand owner's advertising effort may affect the likelihood of a seller engaging in deceptive practices for several reasons. First, advertising effort is a signal of product quality for experience goods (Nelson, 1970), and lower-quality products are more often the ones observed buying fake reviews (He et al., 2022). Advertising is an investment that some brands make to help build brand. Second, manufacturers who invest in building and defending their brand should be more likely to monitor and govern the actions of the sellers of their products. Third, the sellers of products with lower brand advertising support may be tempted to commit review manipulations to overcome the lower manufacturer support for the products they sell. In the context of hotels, Hollenbeck, Moorthy, and Proserpio (2019) found that advertising spending by chains (compared to independents) generally does not respond to their TripAdvisor ratings, which they interpret as being due to a strong, well-known brand that provides some immunity to reviews. Fourth, depending on the message, advertising can equip consumers with relevant knowledge to help them assess the legitimacy of the claims made in an online product review and uncover possible deception. This lowers the effectiveness of efforts to benefit from creating manipulative reviews. Collectively, we argue that there is a negative association between high brand advertising spending and the likelihood that a review of that brand is perceived as suspicious.

2.3. Price

Price affects the incidence of suspicious reviews in two ways. First, high prices are a signal because, from Spence's (1973) conditions, they result in a loss of revenue (costly to signal), and a low-quality seller incurs more costs to mimic a high-quality seller (Milgrom & Roberts, 1986).

Second, price affects the level of consumer involvement and, thus, the amount of attention consumers give to reviews. When consumers encounter some form of communication, such as an online review, they can process it using varying levels of thought or cognitive effort called elaboration (Petty & Cacioppo, 1986). The result is a continuum anchored by two major routes that are useful for understanding how consumers process online reviews and their perceptions of deception. The central route (elaborate processing) entails scrutiny and thoughtful processing, leading to a higher likelihood of detecting review manipulation. The peripheral route (cognitive shortcuts) entails using cues and heuristics that should leave the consumer more susceptible to failing to detect deception. Factors that contribute to elaboration include motivation, ability, and opportunity. Consumers are more likely to be susceptible to deception in online product reviews when they lack the motivation to engage in effortful thinking and/or the ability (relevant knowledge) to process the review's content.

Personal involvement is an important determinant of elaboration motivation. Consumers with greater personal involvement in a purchase decision will be more likely to devote cognitive effort to processing the communication (online reviews). They will also be more likely to rely on sources of information beyond the reviews provided on the seller's platform. Indeed, for high-involvement products, Gu, Park, and Konana (2012) found that reviews posted on a seller's (Amazon) website (called "internal word-of-mouth [WOM]") have a limited influence on sales compared to when these come from independent (e.g., Epinions) sources (called "external WOM").

While product category involvement is a consumer-based concept, a product/brand can also be classified as having high or low involvement based on some of its characteristics (Hoyer, MacInnis, & Pieters, 2017). For example, high prices, technological complexity, and longevity are associated with higher levels of consumer involvement in a purchase decision. Perceived risk is a key determinant in this regard. Consumers deal with perceived risk by being more purposeful and comprehensive when gathering information. When the price of a product is high, consumer involvement is high, and when consumers' engagement with products is higher, they are more likely to scrutinize the products' various aspects, including their reviews. Thus, manipulative reviews of high-priced products may be less effective, reducing the motivation to publish them.

In summary, signaling and consumer involvement together suggest that higher prices are associated with a lower likelihood that a review will be perceived as suspicious.

2.4. Sales rank

Sales rank is an important outcome measure that provides an approximate indication of how popular a product is compared to others in the same category (Prasad et al., 2017). Fake reviews are recognized as producing favorable immediate impacts on star ratings, reviews, and sales rank (He et al., 2022). Firms with lower sales have an incentive to seek a boost from promotional reviews. We argue that a higher sales rank (lower sales; further from #1) is a signal to consumers to be more attuned to the possibility of review manipulation.

2.5. Conceptual model

Fig. 1 illustrates our conceptual model. We expect that the effects will be largely for positive reviews, not for negative ones. He et al. (2022) observed that Facebook groups seeking people to write fake reviews exclusively sought positive product reviews, not fake negative reviews applied to competitors. Our product context is different from consumer services, such as hotels (Mayzlin et al., 2014) and restaurants (Luca & Zervas, 2016), where negative reviews of competitor's products have been found, likely due to the ease of identifying a specific direct, local competitor.

2.6. Control variables

Product category. A technology product usually requires a high level of consumer knowledge, which subsequently demands that consumers research its features, functions, and related usages. Consumers' expected scrutiny may be a caveat for sellers who are tempted to use manipulative reviews, as consumers may be cautious when examining reviews due to the increased importance of their purchase decisions. For these reasons, sellers may be less likely to use manipulative reviews in technology product categories. We expect that a product from the technology sector compared to non-technology sectors has fewer reviews consumers view as suspicious.

Review-level factors. We included several review-level controls that have been found to be associated with fake reviews: review length, star rating, and the emotional content of the text when it is present in a review.

2.7. Contextual conditions affecting the effect of the price signal

Stronger brands are less responsive to price. We expect that the negative effect of price on the likelihood of being a suspicious review will be attenuated (less negative) when perceived brand strength is higher. We also expect the negative effect of price on the likelihood of being a suspicious review to be attenuated in the technology sector due to customers' tendency to seek more in-depth knowledge about the product's features, regardless of how expensive the product may be.

2.8. Empirical investigation

In sum, our problem is to investigate whether perceived brand strength, brand advertising effort, price, and sales rank are predictors of whether a review is suspicious to consumers. We require a set of reviews categorized as suspicious or not. Ideally this would be done by showing people all the online reviews and asking them to rate which ones seem fake. The analysis would then draw conclusions based on the cues they used. This is unfeasible due to the number of observations in our data. So, we examine consumer perceptions of review authenticity as perceived by machine learning algorithms trained on human subjects.

3. Data and labeling reviews as "suspicious (or not)

Amazon review data were obtained from the Stanford Network Analysis Project group at Stanford University (McAuley & Leskovec, 2013) (the data are now stored at University of California San Diego). This data spanned a period of 18 years, ending in March 2013. The data had product, review, and user attributes, including a unique product identifier (Amazon Standard Identification Number [ASIN]), product title, price, unique user identifier, user (reviewer) name, helpfulness score,

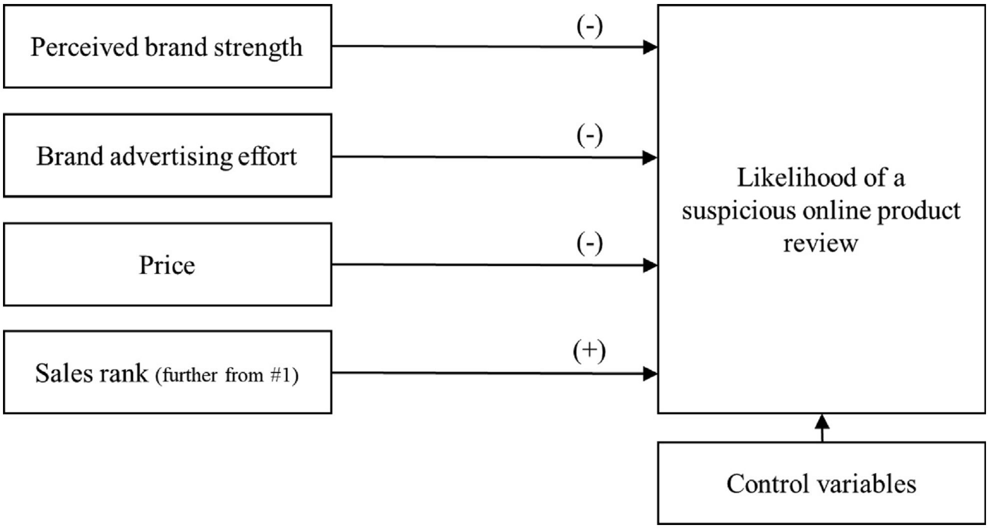


Fig. 1. Conceptual model. *Note.* Signs in parentheses indicate the expected direction of the effect.

review score, review time, review summary, and review text. We also obtained Amazon metadata that included ASIN, product title, price, brand, and co-purchasing links. It is the seller (not the product manufacturer) on Amazon who fills in the product title, brand, etc. Since the brand information was indexed with a unique product identifier, we could link the brand information to the product-level review data. The brand information was also used to facilitate the merging of a measure of perceived brand strength and brand advertising expenditures.

Our starting point was reviews from 16 product categories: arts, beauty, cell phones and accessories, clothing and accessories, electronics, gourmet and foods, health, home and kitchen, industrial and scientific, jewelry, shoes, software, sports and outdoor, tools and home improvement, toys and games, and watches.[3]

3.1. Initial preprocessing of Amazon data

We performed three initial preprocessing steps. First, we removed unknown users and their reviews. Second, we identified sets of duplicate products in the dataset. Amazon.com maintains duplicate product listings, which are essentially the same products with some minor variations, such as color or size, and replicates the same reviews across the entire set of products. Using identical reviews and unique identifiers, we removed duplicate products (approximately 7% of the products) and kept only one representative product per set. Third, we retained only those products that had metadata files. There were several instances of products with reviews in the reviews file but no metadata file. We examined a sample of these, and many were generic products (e.g., from the Gourmet and Foods category: "Ground Decaf Irish Creme 8 Oz," "Filled Red Raspberries," and "Sea Salt, 1 lb."). Fourth, we removed unpopular products (those with fewer than three reviews) and inactive reviewers (those with less than three reviews) so that spammer score variables from a behavioral spam detection algorithm could be appended.

3.2. Labeling as suspicious versus non-suspicious

Ideally, human evaluators would categorize the entire set of reviews (and reviewers) as a suspicious review versus a non-suspicious review (and an incredulous reviewer versus a genuine reviewer), but this is unfeasible due to the number of observations in our data. Instead, human evaluators assess a subset of the reviews as suspicious (or not), which forms a 'ground truth' for a semi-supervised machine-learning algorithm to classify the remaining reviews.

The relatively low incidence of suspicious reviews presents a challenge if we are to create the training set for the human evaluators using a randomly selected sample from all reviews (and reviewers). As such we first use an established spam detection algorithm (Lim et al., 2010) (an automated procedure with no human evaluation) to identify reviewers and their reviews that are potentially spam. The algorithm generates four metrics (spammer detection scores) for a reviewer, each describing various behaviors (e.g., writing many reviews in a short period of time or writing multiple reviews for the same product) that characterizes spammers (see Web Appendix Table A1b for details). The four spammer scores inform a sampling

[3] We did not consider books, music, and videos (hedonic products) as they do not need a brand name entered according to Amazon's policy or they are non-branded. Further, almost all of them never advertise. We checked a sample of these products, and none were in the Kantar database containing brand advertising expenditures.

Table 1
Validating the behavioral features with machine learning classifiers.

Predicted (n = 32)	Provided by Human Evaluators					
	Binary logistic		Naïve Bayes		Random forest	
	Suspicious	Not Suspicious	Suspicious	Not Suspicious	Suspicious	Not Suspicious
Suspicious	17	1	14	0	18	0
Not Suspicious	2	12	5	13	1	13
Accuracy	90.6%		84.4%		96.9%	
Precision	94.4%		100%		100%	
Recall	89.4%		73.7%		94.7%	
F1 Score	91.8%		84.9%		97.3%	

procedure was used to select a small number of reviewers and their reviews to be evaluated and labeled by human annotators. This is the training sample.

Next, using the training sample we created ground truths with human annotators who classified reviewers and their reviews as suspicious or not suspicious. Using helpfulness scores as a benchmark, we evaluated the ground truths using the normalized discount cumulative gain (NDCG) method. The NDCG scores assigned by humans consistently surpassed the baseline model based on the helpfulness scores (See Web Appendix A1.1 for the details of the human annotation process and the performance with NCDG).

Finally, we labeled the remaining reviewers and their reviews as suspicious or not suspicious using a semi-supervised method that employed both the labeled data from the human annotators and the unlabeled data to build a classifier for the unlabeled portion of the observations. The semi-supervised classifier assumes that the underlying structure in the ground truths (the labeled data from human annotators) is also the dominant pattern in the unlabeled data and utilizes an expectation–maximization algorithm to obtain maximum likelihood estimates of the model (the spammer detection scores are the variables) parameters and classifications for unlabeled observations that are updated iteratively (See Web Appendix A1.2 for the details of the semi-supervised classification procedure).

3.3. Validating the full set of labels

We then evaluated the full set of (suspicious versus non-suspicious) labels created by the model-based semi-supervised classification method in two ways. First, we assessed the accuracy of these labels using the behavioral features provided by the spam detection algorithm as predictors. We employed three machine learning classifiers (support vector machine, random forest, and logistic model), and they showed superior to decent predictive performance, where the misclassification rates ranged from 1.03 to 14.55% (see Web Appendix A1.3 for the details of the evaluation procedure and its performance).

Second, we tested the performance of the behavioral features from the spam detection algorithm using new (suspicious versus non-suspicious) labels acquired separately from human evaluators. To create the new labels from the Amazon dataset, we randomly selected 25 reviewers whose reviews were associated with suspicious reviews and 25 reviewers whose reviews were not. The final data for the human evaluation process comprised 55 suspicious reviews by 25 reviewers and 36 non-suspicious reviews by another 25 reviewers. Three human evaluators screened the 91 (55 + 36) reviews. They scored each review on a five-point Likert scale: 1 = non-spammer, 2 = slightly suspicious, 3 = somewhat suspicious, 4 = highly suspicious, and 5 = spammer. We coded the first two levels as not suspicious (coded as 0) and the remaining levels as suspicious (coded as 1). We then assigned a label to each reviewer using majority voting and used it as our ground truth.

We evaluated the ground truths provided by human evaluators as an outcome variable and the behavioral features from the spam detection algorithm as predictors. We split the data into a training set (65%) and a test set (35%) and employed a binary logistic, a naïve Bayes, and a random forest classifier. All three machine learning classifiers provided reasonably high and consistent predictive power with the test set (Table 1). Random forest showed the highest accuracy (96.9%), while naïve Bayes provided the lowest accuracy (84.4%). We further examined the correlation between the labels with the semi-supervised classifier and the labels provided by human evaluators for the 91 reviews; the correlation between these two types of labels was 0.82.

3.4. Validation of the labels in an independent setting

We validated the performance of the spam detection algorithm used to generate the spammer detection scores in an *independent setting* to see if the algorithm had the capability to differentiate fake reviews that were artificially created by human workers from Amazon reviews (which are a combination of real and fake reviews) in our dataset. We compared the characteristics of suspicious reviews (our labels) with fake reviews created by human workers.

For the tests, we created fake reviews ourselves so that we knew for certain that those reviews were fake.[4] We randomly selected 32 products from the Amazon data. Using their product images and descriptions, nine research assistants (RAs) who were blind to our project created 288 (9 × 32) fake reviews. We referred to these as known (RA-created) fake reviews. The 32 products had a total of 1,216 Amazon reviews in our dataset.[5]

We first compared the performance of the algorithm to differentiate known (RA-created) fake reviews from Amazon reviews in a combined dataset of 1,504 reviews (288 + 1,216). Overall, the algorithm returned higher spamming scores, which are indicative of a fake review, for the known (RA-created) fake reviews than for the Amazon reviews (see Web Appendix Table A1g). Second, we compared the performance of a human rater to differentiate a sample of known (RA-created) fake reviews from Amazon reviews. The human rater showed a decent level of accuracy (78%).

Next, we compared the content features of (a) the known (RA-created) fake reviews versus the Amazon reviews for the 32 products and (b) the Amazon reviews labeled as suspicious versus not suspicious in our dataset. Prior research has identified several content features of fake reviews. We examined (1) words per sentence (a smaller number of words per sentence for a fake review), (2) level of details (fewer details for a fake review), (3) number of self-references (fewer self-references for a fake review), (4) emotional magnitude (more emotional exaggeration for a fake review), and 5) number of tentative words (more tentative words for a fake review) (see Web Appendix 1, Tables A1h and A1i for the hypotheses and measures related to the content variables). We conducted a series of analyses (see Web Appendix Tables A1j and A1k for a means comparison, and Table A1l for a binary logistic analysis) and observed (a) that the known (RA-created) fake reviews are differentiated from the Amazon reviews, and (b) that reviews labeled as suspicious in the analysis dataset are similarly differentiated from those labeled as not suspicious. The results are consistent across the two datasets, which supports the face validity of the suspicious review label used in the analyses.

3.5. Merging brand advertising expenditure data

Kantar Media estimates advertising expenditures for more than three million brands across 20 media platforms. Our merging process retained only those brands that appeared in both the Amazon data and the Kanter data. Thus, we dropped many small brands seen on Amazon that have very little, if any, advertising, and thus do not appear in the Kantar database. The merging process used to align the datasets resulted in a dataset containing 1,461 brands, 7,944 products, and 49,508 reviews. Thus, our sample after this merger is branded products that have demonstrated that they have the resources and potential desire to advertise, as evidenced by their inclusion in the Kantar Media database. Some examples from the Tools and Home Improvement category are Black & Decker, Bosch, Delta, Forrest, Freud, Klein, Makita, Milwaukee, Skil, and Stanley.

3.6. Merging perceived brand strength data

Lovett, et al. (2014) published a dataset that includes quarterly Young & Rubicam (Y&R) Brand Asset Valuator pillars measures for 629 brands. The majority of these are national (United States) brands with a strong or exclusive consumer market focus. The Amazon data contains many smaller brands and brands focused on business markets that are not included in the Y&R data. For example, of the 10 brands listed above from Amazon's Tools and Home Improvement category, only Black & Decker appears in the Y&R data. This merging process resulted in 115 brands, 1,411 products, and 8,960 reviews. Web Appendix Table A2 lists the brands from the Y&R data that are also present in the Amazon data.

4. Variable extractions and descriptive statistics

4.1. Variable extractions

Perceived brand strength (BrandStrength): From the Y&R data published by Lovett et al. (2014), for each brand, we selected the most recent value for brand strength (Differentiation × Relevance) available at the end of their observation window (Q2 2010).

Brand advertising expenditure (BrandAdExp): Brand-level advertising spending in dollars from the Kantar Advertising Insights (a precursor to Vivvix) database.

Price (Price) and sales rank (SalesRank): The metadata contains the price and sales rank associated with an ASIN. Amazon determines sales rank using sales in relation to other products in the same product category.

Technology (TechProduct): Electronics, cellphones, and software are clustered as a tech group (coded as 1), and the other product categories are classified as a non-tech group.

Word count per review (WordCount): The number of words in a review.

Star rating (StarRating): At the review level, an integer ranging from one to five.

[4] We thank the Area Editor for suggesting this. This is similar in spirit to Ott et al. (2011) who had MTurkers create (known) fake reviews for the 20 most popular hotels on TripAdvisor in the Chicago area. Like the MTurkers used by Ott et al., our research assistants are likely not as skilled at writing a (known) fake review compared to someone with more experience doing it and/or who is motivated by the potential for economic gain from doing it.

[5] About 4% (48 observations out of 1,216 observations) of the Amazon review is the data labeled as suspicious.

Emotional characteristics of reviews: After preprocessing the text data through cleaning and stemming, we extracted eight primary emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust) by employing a text mining system with a dictionary-based sentiment approach that extracts emotionality from the text. The term frequency of an emotion was measured as its incidence divided by the document length (the total number of terms) for normalization.

We (k-means) clustered the emotions into groups to reduce multicollinearity (e.g., the correlation between joy and anticipation was 0.60). Three clusters reasonably characterize the reviews: 1) a group whose cluster means are high on negative emotions (Cluster 1, *EmotionNegative*, 2) a group whose cluster means are slightly below zero for all eight emotions (Cluster 2, *EmotionNeutral*), and 3) a group whose cluster means are high on positive emotions (Cluster 3, *EmotionPositive*). We created a vector of indicator variables to code which cluster a review was assigned to. Web Appendix, Table A3a presents the means of the basic emotion variables for each cluster.

4.2. Descriptive statistics

Table 2 presents the descriptive statistics, including the mean, standard deviation, minimum value, maximum value, and skewness. Pairwise correlations for the metric variables after standardization are reported in Web Appendix 3, Table A3b.

We examined the distribution of star ratings. They are positively skewed, following the characteristic *J*-shape (Aral, 2013; Brandes, Godes, & Mayzlin, 2022) found in online user reviews (Fig. 2). Most reviews (60%) had the highest star rating of 5. The least frequent star rating (5%) was two. Most products had a high average rating: 49% of products had an average rating between 4.5 and 5. Most users awarded very high scores: 59% gave an average star rating between 4.5 and 5.

To compare the distribution of suspicious reviews with the distribution of reviews not deemed suspicious, we created the distribution for each group separately ([b] in Fig. 2). The distribution is more skewed for suspicious reviews, revealing that suspicious reviews include more extreme ratings (more one- and five-star ratings) than reviews that are not labeled suspicious.[6] Thus, model-free evidence implies that extreme ratings may be significant predictors of suspicious reviews.

Next, we coded our focal variables of interest, describing signals of review reliability—*BrandAdExp*, *BrandStrength*, *Price*, and *SalesRank*—into indicators of high versus low using their means. We created a bar chart showing the proportion of suspicious reviews for each. Fig. 3 shows that suspicious reviews appear more often for branded products with lower perceived brand strength, lower advertising expenditures, and lower prices.

5. Analysis strategy and results

5.1. Model

We predict that a review for a branded product that has a weaker perceived brand strength, a lower brand advertising effort, a lower price, or a higher sales rank (further from #1 in the product category) is more likely be viewed as suspicious by consumers. We estimated the following logistic regression model to describe the relationship between being a suspicious review and brand-level, product-level, and review-level characteristics:

$$Suspicious_{ijt} = W_{ijt}\beta^{w}_{ijt} + X_{jt}\beta^{x}_{jt} + Y_{ijt}\beta^{y}_{ijt} + \varepsilon_{ijt,} \tag{1}$$

$Suspicious_{ijt}$ indicates whether review by user *i* on product *j* at time *t* is a suspicious review or not (a binary indicator); W_{ijt} is a vector of brand-level variables, which are perceived brand strength and brand advertising expenditure;X_{jt} is a vector of product-level variables, which are price, sales rank, and a binary indicator of whether a product is a technology product; and Y_{ijt} is a vector of review-level variables, including word count per review, star rating, and an emotion group (reference level = neutral).

5.2. Results for perceived brand strength, brand advertising expenditure, price, sales rank, and the control variables

All continuous variables were standardized. Without scaling, the models did not converge. We employed a binary logistic model (suspicious versus not suspicious) for the analysis. The estimated results are reported in Table 3. We first ran a baseline model (Baseline) that included perceived brand strength, brand advertising expenditure, and product-level variables (price, sales rank, and technology product indicator). Next, we included the review-level control variables (Model 1). Finally, (Model 2), we incorporated the interaction effects between price and perceived brand strength, between price and brand advertising expenditure, and between price and product type (technology product versus non-technology product). The rightmost column reports the marginal effects for each variable in Model 2.

We found a negative association between perceived brand strength and the likelihood of being a suspicious review. A strong brand signals to the market that the brand is less likely to engage in review manipulation. These brands may be more likely to monitor reviews posted by the sellers of the brand. Consumers can expect that sellers of brands with a weaker brand may seek to overcome this deficiency through review manipulation. Economically, for a one-unit increase in standardized perceived brand strength, the probability of having a suspicious review decreases by 0.62%.

[6] The confidence intervals around the point estimates for suspicious reviews are larger than those for nonsuspicious reviews given their smaller sample size.

Table 2
Descriptive statistics (n = 8,960).

Variables	Mean	Std. dev.	Min.	Max.	Skewness
Signals of online review (un) reliability					
BrandStrength	1.76	0.51	0.68	4.24	1.13
BrandAdExp	41,211	47,441	0.2	618,824	3.63
Price	47.88	90.36	0.01	999	5.71
SalesRank	89,775	148,355	6	1,488,902	2.52
Control variables: Product type					
TechProduct	0.22	0.41	0	1	1.35
Control variables: Review-level factors					
WordCount	75.80	39.36	7	151	0.11
StarRating	4.21	1.20	1	5	−1.49
EmotionNegative	0.17	0.38	0	1	1.71
EmotionNeutral	0.66	0.47	0	1	−1.56
EmotionPositive	0.17	0.37	0	1	1.17

Note: $999.99 is the observed maximum value for price for all products in the entire Amazon dataset. We suspect that this might be the maximum value allowed in this data field. There are 12 observations in the analysis dataset with this value. Our substantive results do not change if these observations are excluded.

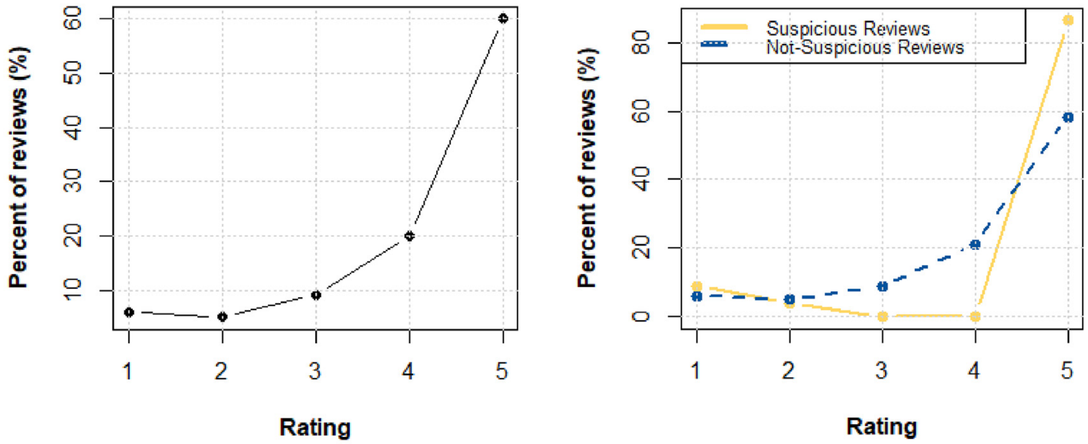
Table 3
Results: Signals of online review (un)reliability—perceived brand strength, brand advertising effort, price, and sales rank.

Variables	Exp. Sign	(1) Baseline	(2) Model 1	(3) Model 2	(4) Marginal effect
Dependent variable: Is the customer review suspicious (y/n)?					
Intercept		−3.14***(0.06)	−4.49***(0.28)	−4.64***(0.31)	
Signals of online review (un)reliability					
Scale(*BrandStrength*)	(−)	−0.11#(0.07)	−0.12#(0.07)	−0.16#(0.08)	−0.0062
Scale(*BrandAdExp*)	(−)	−0.28**(0.09)	−0.28**(0.09)	−0.32**(0.10)	−0.0126
Scale(*Price*)	(−)	−0.12(0.08)	−0.13#(0.09)	−0.60*(0.24)	−0.0243
Scale(*SalesRank*)	(+)	0.32***(0.04)	0.34***(0.04)	0.35***(0.04)	0.0138
Control variables: Product type					
TechProduct	(−)	−0.83***(0.19)	−0.80***(0.19)	−0.83***(0.20)	−0.0327
Control variables: Review-level factors					
Scale(*WordCount*)		–	0.02(0.06)	0.03(0.06)	0.0013
StarRating: Numeric			0.31***(0.06)	0.31***(0.06)	0.0122
EmotionNegative	(+)	–	0.12(0.15)	0.14(0.16)	0.0052
EmotionNeutral (ref)		–	–	–	
EmotionPositive	(+)	–	0.19(0.15)	0.20(0.19)	0.0083
Interactions with the price signal					
Scale(*Price*BrandStrength*)	(+)	–	–	0.47#(0.27)	0.0187
Scale(*Price*BrandAdExp*)	(+)	–	–	0.18(0.15)	0.0073
Scale(*Price*)**TechProduct*	(+)	–	–	−0.05(0.22)	−0.0022
n		8,960	8,960	8,960	
AIC		3,008.3	2,980.2	2,979.2	

# $p < 0.05$; *$p < 0.01$; **$p < 0.001$; ***$p < 0.0001$.
Note: Variables prefaced with the word ‘Scale’ have been standardized.
Note: A positive coefficient indicates a greater likelihood of being a suspicious review.

To investigate the managerial significance of the results, we ran simulations, setting each metric variable at its mean, and calculated the predicted probability of a review being a suspicious review when a focal variable was increased to two standard deviations above its mean (Web Appendix 4). The predicted probability of being a suspicious review declines by 27% when *BrandStrength* increases from its mean to two standard deviations above its mean. The change was not sensitive to different values for *TechProduct* and the emotion indicators (Web Appendix, Table A4b).

We found that, all else being equal, the level of brand advertising effort was negatively related to the likelihood that a review was suspicious. This aligns with our expectation that a lower brand advertising effort is associated with more suspicious reviews. We speculate that when a seller feels that a product they are selling does not have enough marketing support, they may feel a need to find an alternative method to promote it. Sellers tend to avoid this behavior when the branded product they are selling is supported by a higher brand advertising effort. Manufacturers can reliably use brand advertising efforts as a signal of lower review suspiciousness. Economically, the results show that the probability of a suspicious review decreases by 1.3% for a one-unit increase in standardized brand advertising expenditure. The predicted probability of a review being suspicious declines 47% when *BrandAdExp* increases from its mean to two standard deviations above its mean (Web Appendix 4), all else held at their means.



(a) Distribution of star ratings

(b) Distribution of star ratings: suspicious vs. not-suspicious reviews

Fig. 2. Percentage of reviews by star rating. *Note.* (Plot a) one-star rating = 6%; two-star rating = 5%; three-star rating = 9%; four-star rating = 20%; five-star rating = 60%. *Note.* (Plot b) Suspicious reviews: one-star rating = 9%; two-star rating = 4%; three-star rating = 0%; four-star rating = 0.005%; five-star rating = 87%; Not suspicious reviews: one-star rating = 6%; two-star rating = 5%; three-star rating = 9%; four-star rating = 21%; five-star rating = 59%.

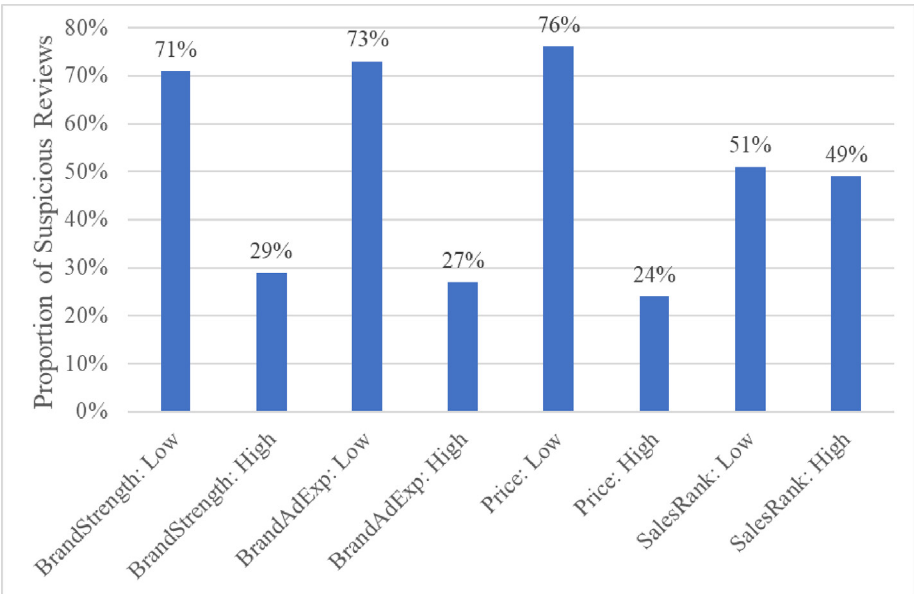


Fig. 3. Percentage of suspicious reviews for each of the focal predictors. *Note.* 'SalesRank: High' means further from #1 in the product category (i.e., lower sales).

The results reveal that a lower price is significantly related to an increase in the likelihood of a review being viewed as suspicious. The estimated results presented in Table 3 show a negative relationship for price. This result suggests that sellers of high-priced products are less likely to have suspicious reviews. The economic interpretation of these findings is that the probability of having a suspicious review decreases by 2.4% for a one-unit increase in the standardized price of a product. Finding that price has the highest responsiveness of the four signaling variables conflicts with Dawar and Parker's (1994) argument that the relative usefulness of price as a signal is lower than that of signals with higher specificity, such as brand and advertising. The predicted probability of a review being suspicious declines 70% when *Price* goes from its mean to two standard deviations above its mean (Web Appendix, Table A4a).

Table 4
Results with a probit model: Signals of online review (un)reliability—perceived brand strength, brand advertising effort, price, and sales rank.

Variables	Exp. Sign	(1) Baseline	(2) Model 1	(3) Model 2	(4) Marginal effect
Dependent variable: Is the customer review suspicious (y/n)?					
Intercept		−1.38***(0.09)	−4.49***(0.28)	−4.64***(0.31)	
Signals of online review (un)reliability					
Scale(*BrandStrength*)	(−)	−0.05(0.03)	−0.05(0.03)	−0.08*(0.04)	−0.0068
Scale(*BrandAdExp*)	(−)	−0.09*(0.04)	−0.09*(0.04)	−0.11**(0.04)	−0.0094
Scale(*Price*)	(−)	−0.05(0.04)	−0.05(0.04)	−0.28#(0.14)	−0.0267
Scale(*SalesRank*)	(+)	0.16***(0.02)	0.17***(0.02)	0.17***(0.02)	0.0149
Control variables: Product type					
TechProduct	(−)	−0.34***(0.08)	−0.32***(0.08)	−0.33***(0.08)	−0.0281
Control variables: Review-level factors					
Scale(*WordCount*)		–	0.01(0.03)	0.01(0.03)	0.0011
StarRating: Numeric			0.11***(0.02)	0.12***(0.02)	0.0099
EmotionNegative	(+)	–	0.12(0.09)	0.12(0.09)	0.0098
EmotionNeutral (ref)		–	–	–	
EmotionPositive	(+)	–	0.05(0.07)	0.06(0.08)	0.0056
Interactions with the price signal					
Scale(*Price*BrandStrength*)	(+)	–	–	0.27*(0.13)	0.0232
Scale(*Price*BrandAdExp*)	(+)	–	–	0.06(0.06)	0.0055
Scale(*Price*)**TechProduct*	(+)	–	–	−0.04(0.09)	−0.0033
n		8,960	8,960	8,960	
AIC		3,010.7	2,990.7	2,987.8	

# $p < 0.05$; *$p < 0.01$; **$p < 0.001$; ***$p < 0.0001$.
Note: Variables prefaced with the word 'Scale' have been standardized.
Note: A positive coefficient indicates a greater likelihood of being a suspicious review.

A review is more likely to be viewed as suspicious for products with a higher sales rank (lower sales; further from #1 in the product category). The coefficient of *SalesRank* is positive and significant, which aligns with our prediction. Economically, the results show that for a one-unit increase in the standardized sales rank, the probability of being a suspicious review increases by 1.4%. The predicted probability of a review being suspicious doubles when *SalesRank* increases from its mean to two standard deviations above its mean (Web Appendix 4). We also investigated several interactions involving price and found that the interaction effect of price and perceived brand strength was positive and marginally significant (Table 3, Model 2).

In a robustness check, we employed a probit model specification (Table 4). The substantive findings are consistent with the results reported in Table 3.

5.3. Results for positive and negative reviews analyzed separately

We separated the reviews into a positive (POS: reviews with four or five stars) and a negative (NEG: reviews with one or two stars) group and estimated the models reported in Tables 3 and 4 for each subset. As shown in Fig. 2, 80% of the reviews were positive, and most of the reviews coded as suspicious were positive. This meant that there were few observations with suspicious reviews in the NEG dataset. Therefore, we created a larger dataset by omitting the perceived brand strength variable. The Y&R data were largely national brands and merging them with the Amazon data resulted in many observations being deleted.

Web Appendix, Tables A5a and A5b present the descriptive statistics and correlations for this larger dataset. These are comparable to the subset used in the analysis dataset described above (analogous to Table 2 and Web Appendix, Table A3b). Web Appendix, Table A5c presents the results for the binary logistic model separately for the positive and negative reviews (analogous to Table 3). Web Appendix, Table A5d presents the results for the probit model separately for the positive and negative reviews (analogous to Table 4).

We found that the substantive findings presented in Section 5.2 are robust for positive reviews. The results are also consistent for negative reviews, but with two notable exceptions. First, the coefficient of *BrandAdExp* is no longer statistically different from zero. A brand's advertising efforts are not a reliable signal of suspiciousness for negative reviews. Second, for negative reviews, the interaction of price with the indicator of being a technology product is positive and significant. For technology products, higher prices are associated with a great likelihood of a negative review being a suspicious review.

6. Conclusions

Consumers are often suspicious of the authenticity of online product reviews. The direct influence of online reviews on purchasing decisions makes opinion-sharing platforms attractive targets for review fraud.

We study suspicious reviews, reviews that consumers are suspicious of being fake. Suspiciousness can lead to a belief that a brand is of lower quality (Zhuang et al., 2018), cause consumers to underweight reviews (Munzel, 2015), can negatively affect attitudes toward the brand (Harrison-Walker & Jiang, 2023), hurt the reputation of the opinion-sharing platform, and ultimately have a negative impact on brand sales. It is important to understand consumers' *perceptions* of deception because it is a consumer's perception regarding online reviews that leads to their feelings, opinions, and, ultimately, their actions (Román, Riquelme, & Iacobucci, 2019). We study consumer perceptions of review authenticity as perceived by machine learning algorithms trained on human subjects.

We propose that perceived brand strength, brand advertising effort, price, and sales rank each acts as a signal that helps positive reviews of a brand seem more reliable to consumers. We find support for these factors using Amazon.com reviews of branded products from 16 product categories. We highlight the findings below.

First, stronger perceived brand strength is associated with a lower likelihood that a review is suspicious. A brand must devote resources to building strength, which consumers can use as a signal to gauge the likelihood of deceptive practices. Sellers of weaker brands have an economic incentive to compensate for this shortcoming by engaging in deceptive practices.

Second, higher brand advertising effort measured by advertising expenditures is associated with a lower likelihood that a review is a suspicious review. This finding, particularly for positive reviews, denotes that the sellers' intention to promote what they sell is stronger for brands with lower advertising support and, thus, may lead to a greater desire to employ deceptive methods to boost sales. The effect does not hold for suspicious reviews with defamatory purposes.

Third, our analyses revealed a lower likelihood that a review would be suspicious for products whose price is higher. A possible explanation for this finding is that for products where consumers are more involved in the purchase decision (high-priced products), it may not be easy for a seller to engage in review manipulations because false reviews can be more easily detected by consumers, given their higher level of engagement with the product's features and customer reviews.

Fourth, lower category sales, measured as higher sales rank (further from #1), is associated with a greater likelihood that a review is suspicious. Brands with lower sales in their respective categories have an incentive to seek a boost from a promotional review.

Fifth, suspicious reviews are more likely to occur for products in non-technology sectors. This result suggests that when a customer opinion-sharing platform, such as Amazon, makes policy or acts against merchants engaging in fake review schemes, it needs to consider variations of potentially manipulative reviews depending on the product category.

Sixth, review characteristics may be good references for opinion-sharing platforms that investigate potentially manipulative reviews to act against fake review schemes. The presence of extremely positive or negative emotions better describes suspicious reviews than neutral or weak emotions. This finding is consistent with other contexts, such as storytelling (e.g., Newman et al., 2003). While more extreme reviews tend to have a greater proportion of emotional content (Ullah et al., 2016), we found emotional content to have a considerable effect after accounting for a review's star rating.

Our results have several managerial and academic implications. First, the associations of brand strength, brand advertising effort, price, and sales rank with suspicious reviews can serve as useful references for consumers and for firms whose operations are centered on customer reviews. Firms investigate and seek to regulate fake reviews. The information offered from our empirical analysis regarding the characteristics of suspected manipulative reviews can help the managers of such firms better govern review quality and maintain their platforms' reputations.

Second, we believe that the suspicious review characteristics identified in this study may be beneficial to consumers. Although newer generations are more wary of online reviews, consumers do not have enough knowledge about the trends or features that characterize potentially manipulative reviews. We hope that consumers will become more knowledgeable using the findings of the current research, which show the trends associated with suspicious reviews, such as the higher incidences of suspicious reviews with weaker brand strength or lower brand advertising efforts, or a lower number of suspicious reviews when there is higher product involvement. Consumers empowered by a higher sensibility could be beneficial not only for themselves but also for the general business environment. A business ecosystem that is monitored by consumers with higher sensibility makes e-commerce businesses or online-based inputs more trustworthy and legitimate.

Finally, it is established that small or unknown brands engage in review manipulation in the hotel (Mayzlin et al., 2014) and restaurant (Luca & Zervas, 2016) industries and that they are the ones observed buying promotional reviews on Facebook (He et al., 2022). If there is a financial incentive to engage in review manipulation, this should be done by the sellers of these products. Our study adds to these findings, showing that there are issues with review integrity and perceived review integrity beyond just small or unknown brands.

We highlight that our sample is brands that have an established brand reputation by virtue of their inclusion in the Y&R database, and by their inclusion in the Kantar Advertising Insights database, they have also shown that they have both the resources and desire to advertise. A significant portion of the products sold on Amazon.com come from small or unknown brands with modest, if any, brand reputation and advertising support. These were excluded from the analysis.

Our results are subject to limitations, which suggest opportunities for future research—many of which have implications for data collection. First, we focus on the effects of perceived brand strength, brand advertising effort, price, and sales rank, but there might be other factors that drive suspiciousness, such as how recently a product was launched or characteristics of the seller. Second, there may be further moderating variables that strengthen or weaken the signals of suspiciousness investigated in the current research, such as a brand's positioning. Third, we suspect that there are different reasons for creating negative spam reviews that are not examined in our study. For example, a recent increase in a competitor's market position may motivate review fraud with a defamatory intention against that competitor. Fourth, the spam detection method that we applied, which was based on spammers' behavioral patterns, excluded reviewers who wrote fewer than three reviews, so these reviews were unanalyzed. This has indeed been a major challenge for many spam detection algorithms: they do not examine singleton reviews. Finally, it would be useful to examine factors associated with the incidence of suspicious reviews in settings other than Amazon.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijresmar.2023.06.006.

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